INTRODUCTION

This document provides an update to the Communicable disease alert and response for mass gatherings: key considerations, June 2008. This new version builds on the expertize gained across the global mass gathering (MG) community since this was published. It has also been expanded to cover more than communicable diseases and includes new areas such as legacy, environmental health and different contexts such as unplanned mass gatherings (MGs).

Legacy has been included here as a key element of the planning for MGs. Leaving a viable public health legacy and sustainable improvements in the health infrastructure and capacity should be a key aim of those involved with preparing for any MG. Legacy can include improvements in the health systems in the host country, improvements in health behaviours, and ability to deliver future MGs. It is important that the legacy planning is seen as equal to other areas of planning and both political will and the associated funding should be used to achieve this.

The document was conceived as a resource to support all those responsible for the health needs of individuals attending a MG, and to help them plan their actions. The principles and practices outlined in the document may also provide valuable initial guidance to those involved in planning other aspects of the management of MGs.

The Global Capacity Alert and Response Department (GCAR) of the World Health Organization (WHO) and the network of the WHO Collaborating Centres for Mass Gatherings increasingly provide expertize, support and resources to those planning MGs. This document has been produced to help address these requests, and to provide support to all those involved in the health aspects of planning for MGs, not just those directly employed in the health services.

This document draws on experiences from past MGs that suggest certain common critical factors and preconditions for success, as well as strategic, organizational, and tactical "lessons learned" that can be applied to future gatherings.

Development of this document

This document has been produced as a collaborative project across the network of WHO Collaborating Centres for Mass Gatherings with experts from across the WHO Virtual Interdisciplinary Advisory Group (VIAG). The work was led by a small steering group from WHO and Public Health England as the lead Collaborating Centre.

The steering group and chapter leads were chosen in order to cover the range of specializations included in the document, and because of their experience in planning for and participating in public health at MGs. They are listed in the acknowledgements.

The document will be reviewed after five years, in 2020, taking into account any feedback that has been received in the interim. The structure and content of the document will be reassessed at that time, and any modifications required will be carried out by the steering group in consultation with other relevant experts. A revised version of the document will then be issued.

Background

The decision to host a MG will usually be made well in advance by the key agencies involved, in order to make effective prior planning possible. Such planning is of paramount importance, and addressing public health prevention and response is one of the most important aspects.

MGs are characterized by the concentration of people at a specific location for a specific purpose over a set period of time and which has the potential to strain the planning and response resources of the country or community. The definition is purposefully not linked to the size of the gathering or the number of people (although this obviously has an impact on the assessment of associated risks) because each community has a different a capacity to manage crowds of people, with some systems, for example, airports or market places, managing upwards of 100,000 people on a daily basis with minimal difficulties.

Planning and preparing public health systems and services for managing a MG is a complex procedure: advanced risk assessment and system enhancement are critical to identifying potential public health risks, both natural and manmade, and to preventing, minimizing and responding to public health emergencies.

MGs can place a strain on the local health care system; even the most prepared of events may experience a disaster, which can overwhelm local healthcare systems and their ability to provide an adequate emergency response.

Health services are generally designed to meet routine priorities and demands, and have limited capacity to expand. MGs may put a strain on these systems and so require strengthening of existing services and potentially the introduction of new or enhanced methods for managing disease and other public health risks. These can include surveillance methods, 'Standard Operational Procedures (SOPs), and establishment of a public health response command and coordination structure within and between public health sectors.

About this document

Aim

This document presents the key issues to be considered in the process of setting up and implementing public health alert, response and operational plans for MGs. It sets out methods for assessing the needs of the MG, determining the ability of existing systems to meet those needs, and modifying and strengthening those systems where required. It provides advice about prevention, detection and management of public health incidents, as well as the integration of the full range of public health activities into the MG planning process. It addresses the many different types of MG, provides case studies of the work undertaken for public health activities at various MGs, and supporting papers and additional resources to support the information here.

This document has been developed primarily for public health professionals, as well as key policy makers, planners and executive personnel. In addition to those in the health sector, there are many outside authorities involved in contributing to healthy outcomes at MGs, who will also find this document useful. It is further anticipated that it will be a valuable resource for event promoters and managers, emergency service personnel, government bodies, and any organizations or individuals who contribute to the organization of MGs.

Risk assessments must be regularly reviewed to ensure appropriate and adequate responses. Even well-established and well-organized MGs can turn negative. For example, the 2007 Chicago Marathon in the United States was cancelled mid-race due to unseasonably high temperatures and an inability to guarantee the safety of the 36,000 runners participating.

Public health needs will be determined on the basis of the results of the **strategic risk assessment** for the MG. This is undertaken before the event and requires a thorough examination of potential threats: a threat assessment (including those that seem very unlikely) and a set of standardized questions which help identify the risks to different groups.

In addition to strategic risk assessments a system for **case-based rapid risk assessment** will be required if a significant health event is detected, from the initial alert throughout the duration of response. If an outbreak occurs, and once its aetiology is known, further refinement of the risk assessment may be required.

Strategic risk assessment
This identifies health risks and determines realistic goals for reducing their impact.

MGs may cause an increase in the level of existing risks, or they may pose entirely new risks. The public health risk should be identified and assessed in relation to how the MG will affect the probability of these diseases occurring and spreading in:

- The host nation,
- Nations bordering the host nation, and
- Home nations of participants.

Risk assessment can be broken down into the following steps:

- Identification
- Characterization - evaluation of risks
- Management - decisions on precaution(s).

In a MG the risks may be amplified by a wide range of factors including the venue, locally endemic diseases, the strength of local surveillance systems, the origin of participants, or the intention of groups to deliberately target the event.

In addition to the public health risks it is important to also undertake an analysis of strengths and vulnerabilities in existing health systems, including:

- Surveillance and response: the system's ability to rapidly detect and implement appropriate and commensurate measures to contain / minimize the impact on health; the system's ability to maintain a state of alertness for prolonged periods of time, including through false alarms, fatigue; the system's surge capacity
- Medical services: the local hospital system and its ability to manage an increased number of patients and if necessary to rapidly reach a surge operations in case of an emergency
- Food and water safety: the ability of local infrastructure to ensure safe preparation and delivery of food and water to attendees of MGs.

Risk identification
This is the process of identifying known or potential hazards for the MG. This should include:

- context - type of event
- demographics - participants and / or spectators, both host country and visitors
- normal incidence of public health risks in the host community, including communicable diseases
- environmental factors such as location, access and temperature
- potential importation and / or exportation of communicable diseases
- event additionality required for host country systems / processes
- political and / or media interest.

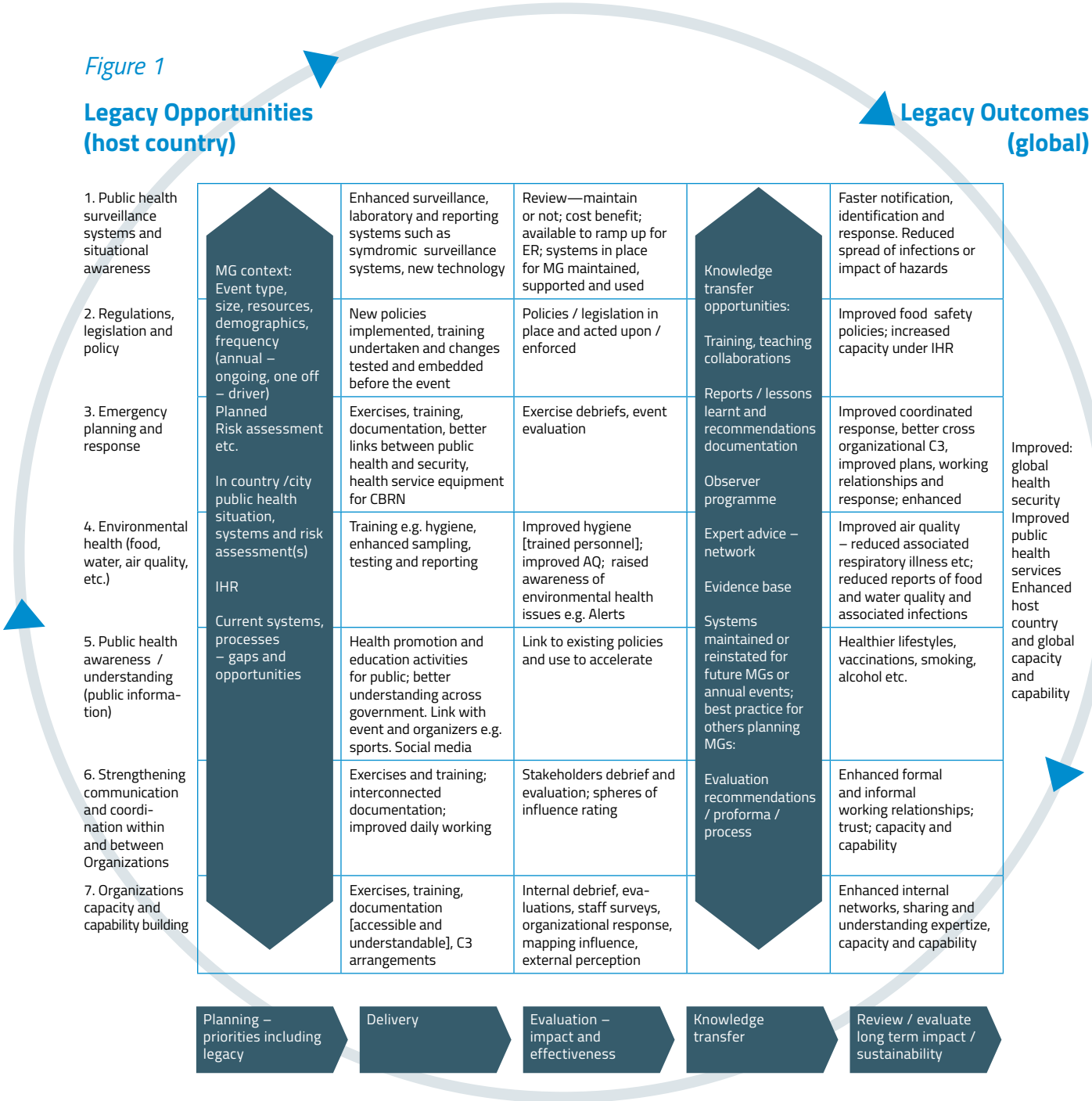
This process can draw on many sources of information including:

- information from previous MG

PRACTICAL SUGGESTIONS AND IMPLICATIONS

A standard health legacy framework (see Figure 1) and evaluation toolkit based on literature and experience can be adapted to fit different contexts and can be used to build the global knowledge and evidence base.

Figure 1



This approach focuses on specific legacy areas and the benefits of a known and sustainable legacy for the host country, local population and global community. It provides information on the key areas identified and the iterative process that could be followed; from identification through delivery, evaluation and dissemination.

During the event

The main focus should be on capturing and documenting the outputs of preparation activities. This includes capturing information and experiences, through documentation such as the formal situation reports and experiences both factual and anecdotal, e.g. blogs, interviews etc.

There should also be a continuous process of evaluation and learning to ensure processes and systems remain effective and that they reflect any changes in expected deliverables.

Observer programmes during an event offer an excellent opportunity to share expertize and experience with those delivering future MGs. Knowledge and experience gained in preparing and running one event can be transferred to others. Health based observer programs can also be linked with observer programmes run by the larger event organizers such as the International Olympic Committee (IOC) and FIFA.

After the event

At the conclusion of a MG, stakeholders should evaluate and document outputs (what happened), successes, lessons learned, and recommendations for the current host, organizing body, and others planning MGs. Any recommendations should be supported with actions on how these will be learnt and embedded across the health environment. The emphasis should be on observing, identifying, describing, recording, and communicating lessons identified. Evaluation should inform legacy through recommendations to the organizations involved, those planning future MGs and the global resource for MG planning.

Documentation of the resources required at all stages is critical, and if at all possible a cost-effectiveness analysis undertaken.

Capturing the learning and experience needs to be done very quickly as those involved rapidly move on to new jobs and the event host organizations can cease to exist. There is also a short window of opportunity and interest in the event from media and politicians.

I. Evaluation

An evaluation of the effectiveness of planning and delivery systems during the event is necessary, both during steady state and any incidents. This should include a post-event assessment of the value and / or effectiveness of any specific interventions or new or enhanced systems set up for the MG: did they meet the objectives for which they were developed?

Evaluation should be done according to a structured framework with outcomes documented to inform the evidence base, particularly looking at the system's usefulness and specific attributes. Information should include the number and type of events and actions taken.

Representatives from different areas of planning and delivery should be included across local, regional and national health organizations as well as undertaking consultations with the different sectors / authorities / agencies involved.

As part of the evaluation it is worth considering:

- Identifying successful practices and how they will be maintained, e.g. surveillance, reporting, microbiology, communications system
- What was and what was not cost-effective and how this can be improved, e.g. setting up new surveillance systems, staffing costs etc. and adapting or eliminating unsuccessful or cost-

SUMMARY OF IHR (2005)

The scope of the IHR (2005) is quite broad and includes the following international public health risks:

- All hazards risk including biological, chemical or radionuclear hazards
- When the disease, agent or mode of transmission is known or unknown, and
- When the hazard is transmitted by:
  - Persons (e.g. SARS, influenza, polio, Ebola)
  - Goods / food / animals (including many zoonotic disease risks)
  - Vectors (e.g. plague, yellow fever)
  - The environment (e.g. releases spills or other contamination).

The IHR create rights and obligations for states that are potentially relevant in the context of MGs, including:

- Notification to WHO of outbreaks which fulfil certain risk assessment criteria
- Verification of these events upon request from WHO
- Rules on application of health measures to international travellers, trade and transportation
- Maintenance or development of core public health capacities for surveillance, assessment, response and communication concerning public health risks and events.

The IHR also mandate a key role for WHO in the surveillance and management of public health events with international consequences. This emphasizes the importance of involvement with WHO when planning for and delivering major MGs.

Through these requirements, the IHR (2005) aims to ensure that all countries have the infrastructure they need to identify, assess, and respond to public health threats around the world.

Public health events occurring during MGs may be reportable to WHO under the IHR, or subject to a verification request from WHO. These include:

- Events in areas with a high population density and therefore having the potential for a high public health impact
- An event occurring in association with an international gathering, resulting in a risk of international travel or trade restrictions
- An event with evidence of local transmission, and a case with a history of attending an international gathering within the previous month.

Designation of PHEIC

The WHO Director General, acting on advice from staff and a specially convened group of international experts external to WHO, termed an IHR Emergency Committee, has the authority to designate an acute public health event as a public health emergency of international concern (PHEIC) according to specified criteria and procedures.

Reporting and assessment

The IHR provide a clear standard framework for identifying how countries notify WHO and what type of events should be reported.

How IHR operates

IHR related communications with WHO are channelled through the country's National IHR Focal Points (NFP). In preparation for the MG, the NFP should be involved in the surveillance reporting and assessment during the MG. In particular if any enhancements to existing surveillance systems or the development of new surveillance systems are planned.

WHO's six Regional Offices have an IHR Contact Point for the countries in their region and should receive all IHR related communications.